Bayesian Statistics Project – A4

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Introduction

Extreme value modeling of ozone data

- Fit a model for **daily extreme ozone values** at each monitoring station.
- The **GEV distribution** is commonly used for this purpose in the literature.
- Incorporate meteorological covariates to assess their influence on ozone extremes.

Data

Sources

- **ARPA Lombardia**: hourly ozone measurements from a regional network of monitoring stations (2018–2024).
- **Open-Meteo**: meteorological data retrieved at station locations.

Processed variables

- weekly maximum ozone concentration.
- weekly meteorological covariates:
  - maximum and minimum temperature,
  - maximum wind speed,
  - total weekly precipitation.

Dataset size

- **32 monitoring stations** across Lombardia.
- weekly observations from **2018 to 2024**.
- Approximately **11,000 station–week observations** after quality control.

Location of Ozone Monitoring Stations

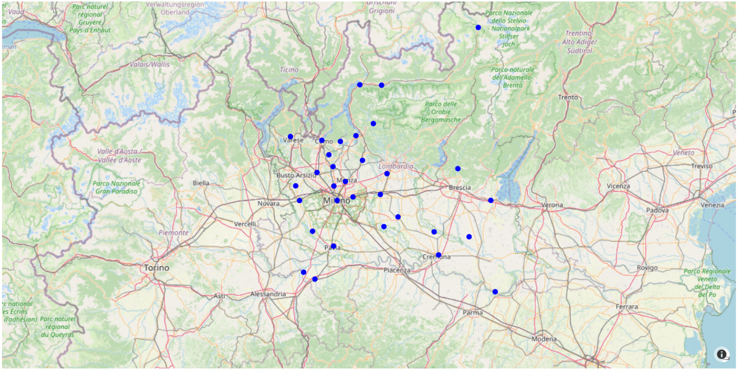


Figure: Location of the sensors

Data exploration

- Ozone maxima are strongly related to temperature,
- Wind speed and precipitation show weaker associations.
- weekly maximum ozone exhibits a strong annual seasonality.
- No clear long-term trend is visible.

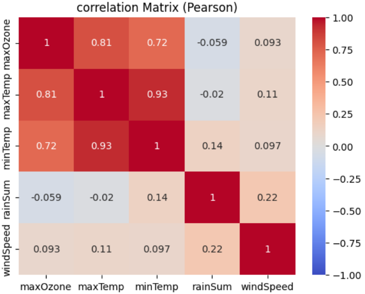


Figure: correlation matrix.

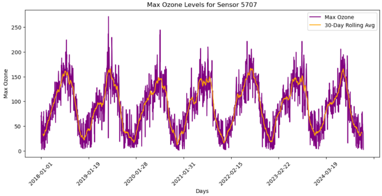


Figure: maximum ozone level for one sensor.

Extreme value modelling: the GEV framework

- The **Generalized Extreme Value (GEV)** distribution describes the behaviour of block maxima (Escarela, 2012).
- It is the standard statistical model for environmental extremes such as ozone maximum.

Tail behaviour

- Controlled by a single shape parameter ξ:
  - $\xi < 0$: bounded upper tail (Weibull),
  - $\xi = 0$: exponential tail (Gumbel),
  - $\xi > 0$: heavy tail (Fréchet).

Limitations

- Classical GEV parameters can be difficult to interpret when the tail is heavy.
- For large ξ, moments may not exist.
- The standard GEV may also impose artificial bounds on the data.

Blended GEV (bGEV): motivation and advantages

- The **blended GEV (bGEV)** extends the classical GEV to improve interpretability and stability.
- It is based on a **quantile-based parametrization** (Castro-Camilo et al., 2022):
  - a location quantile,
  - a quantile range measuring the spread.

Key ideas

- Blend between Gumbel and Fréchet distribution
- The left tail behaves like a Gumbel distribution and the right tail behaves like a Fréchet distribution
- The tail parameter controls only extremeness, not the overall scale.

Why bGEV for ozone?

- bGEV provides realistic tail behaviour while keeping parameters interpretable and stable.
- Well suited for Bayesian inference and covariate-dependent modelling.

Penalized Complexity prior for the tail

We use a **Penalized Complexity (PC) prior** to regularize tail behaviour.

Key principles

- The PC prior shrinks the model toward the simpler Gumbel case ($\xi = 0$) unless the data provide strong evidence otherwise.

Practical constraint

- Values $\xi > 0.5$ imply infinite variance.
- We therefore use a **truncated PC prior** ($0 \leq \xi < 0.5$), ensuring finite mean and variance.

Why this matters

- Prevents unrealistic extreme predictions and improve stability.

Bayesian model

We assume the daily ozone observations are independent and follow a blended GEV with location and scale dependent on covariates:

$$
\begin{aligned}
y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi \;&\overset{\text{ind}}{\sim}\; \mathrm{bGEV}\big(q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi\big), \quad \text{for } i = 1, \dots, S,\, t = 1, \dots, T \\
q_{\alpha,i,t} = \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta}, \qquad s_{\beta,i,t} &= \exp(\mathbf{x}_{i,t}^{\top}\gamma), \quad \text{for } i = 1, \dots, S,\, t = 1, \dots, T \\
\boldsymbol{\beta},\, \boldsymbol{\gamma},\, \xi \;&\sim\; \pi(\boldsymbol{\beta}, \boldsymbol{\gamma}, \xi) = \pi(\boldsymbol{\beta}) \cdot \pi(\boldsymbol{\gamma}) \cdot \pi(\xi), \\
\textit{with } \boldsymbol{\beta} &\sim \mathcal{N}\big(\mathbf{0},\, \Sigma_{\beta}\big), \\
\boldsymbol{\gamma} &\sim \mathcal{N}\big(\mathbf{0},\, \Sigma_{\gamma}\big).
\end{aligned}
$$

where S is the number of monitoring stations and T the number of days we are considering for each location.

Data splitting and inference

- Training data: **weekly** ozone observations from 2018–2023 for **32 stations**.
- Validation data: year 2024
- Covariates have been scaled
- Inference performed using the Laplace approximation implemented in INLA
- $q_{\alpha,i,t}$ is the median
- $s_{\beta,i,t}$ is 50% interquartile range

2 different models tested:

- Model with temporal dynamics.
- Model with with covariates in the spread.

Temporal bGEV Model

Model structure

$$y_{i,t} \sim \mathrm{bGEV}(q_{i,t}, s_{i,t}, \xi)$$

Location (median):

$$q_{i,t} = \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta} + \beta_{\text{year}} \cdot year + f(month)$$

- Linear meteorological effects
- Linear temporal trend (year effect)
- Cyclic seasonal component (second-order random walk)
- Constant spread parameter
- Global tail parameter ξ (P3C prior)

Key Findings – Temporal Model

1. **Temperature is the dominant driver**
   - Maximum temperature: strong positive effect $(+40.4)$
   - Hot conditions strongly increase extreme ozone
3. **Moderate seasonality**
   - Peak in spring–early summer
   - Peak-to-trough difference $\approx 5\ \mu g/m^3$
4. **Light tail behaviour**
   - $\xi \approx 0$ (Gumbel-type tail)
   - No evidence of heavy-tailed extreme events

bGEV Model with Covariates on Spread

Model structure

$$y_{i,t} \sim \mathrm{bGEV}(q_{i,t}, s_{i,t}, \xi)$$

Location (median):

$$q_{i,t} = \mathbf{x}_{i,t}^\top \boldsymbol{\beta}$$

Spread (heteroskedastic):

$$s_{i,t} = \exp(\mathbf{x}_{i,t}^\top \gamma)$$

- No temporal trend or seasonal component
- Median depends on meteorological covariates
- Variability also depends on meteorological covariates
- Global tail parameter ξ (P3C prior)

Key Findings – Heteroskedastic Model

1. **Temperature remains dominant**
   - Maximum temperature: strong positive effect $(+28.7)$

3. **Wind stabilizes extremes**
   - Strong negative effect on spread (-0.166)
   - Wind reduces extreme variability

4. **Rain lowers mean but increases variability**
   - Negative effect on median (-3.58)
   - Positive effect on spread $(+0.029)$

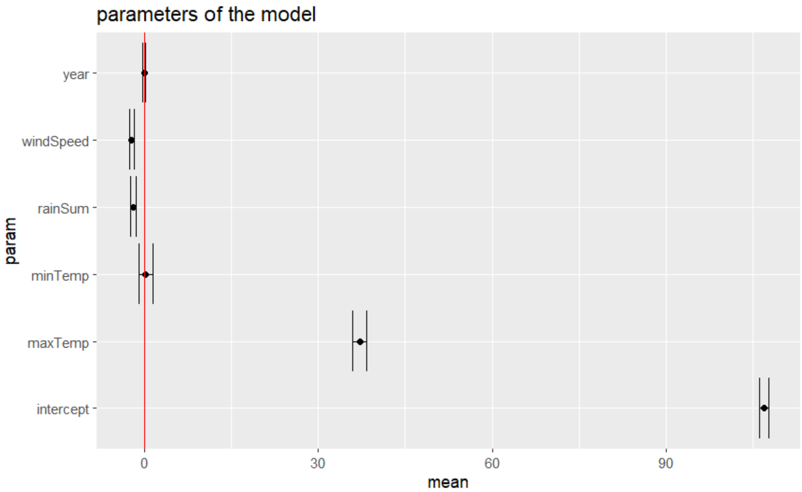
5. **Tail remains light**
   - $\xi \approx 0.0026$
   - Slightly larger than temporal model but still near Gumbel

Model with covariates on spread and temporal components

The model includes temporal components of the first model and some covariates on the spread parameter:

$$
\begin{aligned}
y_{i,t} \mid q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi \;&\overset{\text{ind}}{\sim}\; \mathrm{bGEV}\big(q_{\alpha,i,t},\, s_{\beta,i,t},\, \xi\big)\\
q_{\alpha,i,t} &= \mathbf{x}_{i,t}^{\top}\boldsymbol{\beta} + \beta_{year}\, year + f(month)\\
s_{\beta,i,t} &= \exp(\gamma_0 + \gamma_1 \text{windSpeed} + \gamma_2 \text{RainSum}),\\
\boldsymbol{\beta},\, \boldsymbol{\gamma},\, \xi \;&\sim\; \pi(\boldsymbol{\beta}, \boldsymbol{\gamma}, \xi) = \pi(\boldsymbol{\beta}) \cdot \pi(\boldsymbol{\gamma}) \cdot \pi(\xi),\\
\textit{with } \boldsymbol{\beta} &\sim \mathcal{N}(\mathbf{0},\, 1),\\
\gamma_0 &\sim \text{Log-gamma}(3, 3)\\
\gamma_1, \gamma_2 &\sim \mathcal{N}(0,\, 1).\\
\xi &\sim \text{P3C prior}, \quad \xi \in [0,\, 0.5]
\end{aligned}
$$

Results


parameters of the model
year
windSpeed
rainSum
minTemp
maxTemp
intercept
param
0
30
60
90
mean

Interpretation of location effects

Intercept: $106.92\ \mu g/m^3$

- This Corresponds to the reference high-ozone quantile when meteorological covariates are at their centered values.

Year effect: $-0.05\ \mu g/m^3$ (95% CI: $[-0.30, 0.20]$)

- No clear long-term trend in extreme ozone concentrations.

Minimum temperature: $+0.24$ (95% CI: $[-1.02, 1.50]$)

- not significant

Maximum temperature: $+37.12$

- Dominant meteorological driver.

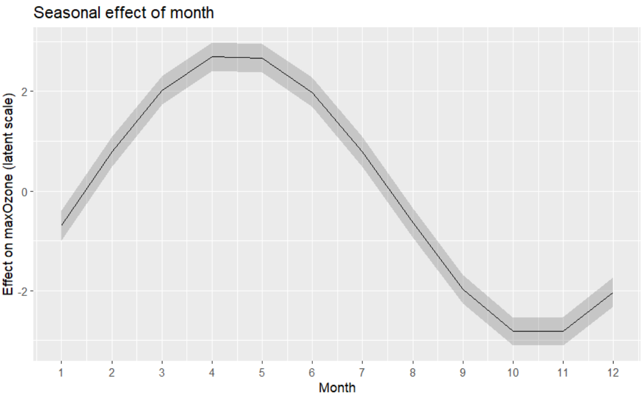
Wind speed: -2.17 (95% CI: $[-2.61, -1.73]$)

- Strong negative effect.
- Increased wind speeds substantially reduce extreme concentrations.

Rain sum: -1.87 (95% CI: $[-2.37, -1.37]$)

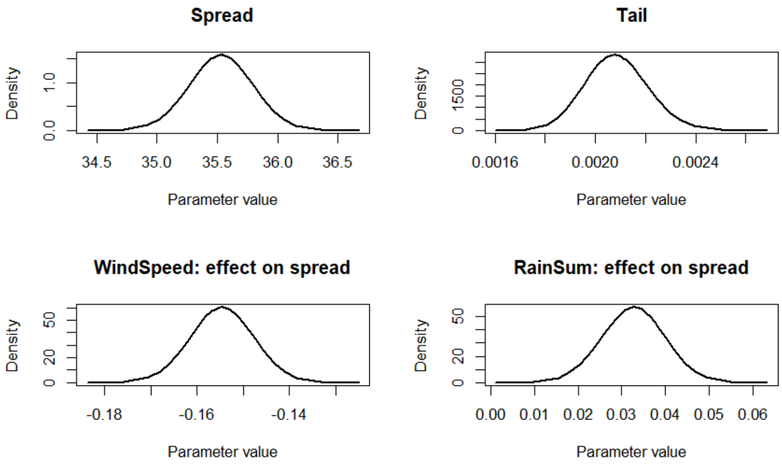
- Precipitation significantly suppresses ozone extremes.

Seasonal effects of hybrid model



Peak zone in spring-early summer, low zone in automn.
Peak-to-trough $= 5.3\ \mu g/m^3$. Moderate effect.

Results


Spread
Density
0.0
1.0
34.5
35.0
35.5
36.0
36.5
Parameter value
Tail
Density
0
1500
0.0016
0.0020
0.0024
Parameter value
WindSpeed: effect on spread
Density
0
20
50
-0.18
-0.16
-0.14
Parameter value
RainSum: effect on spread
Density
0
20
50
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Parameter value

Interpretation of spread and tail effects

Baseline spread: 35.53 $\mu g/m^3$

- Represents the typical variability of extreme ozone concentrations under average meteorological conditions.

Wind speed: -0.15 (strongly significant)

- Wind strongly stabilizes ozone extremes by reducing variability.
- wind limits extreme spikes.

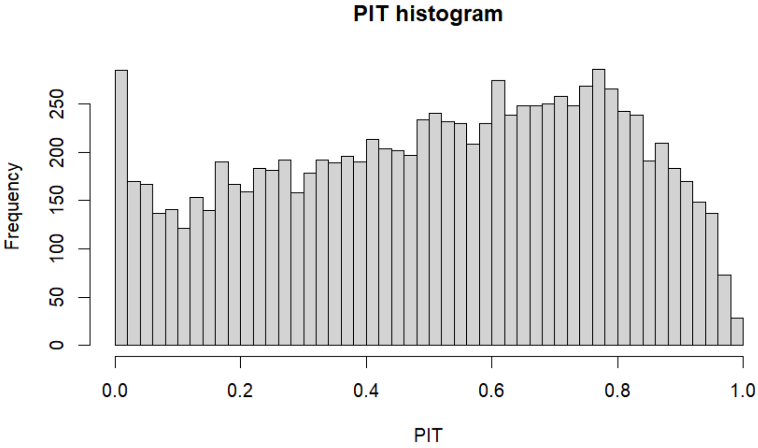
Rain sum: $+0.03$ (significant)

- Rain slightly increases variability despite lowering the mean.

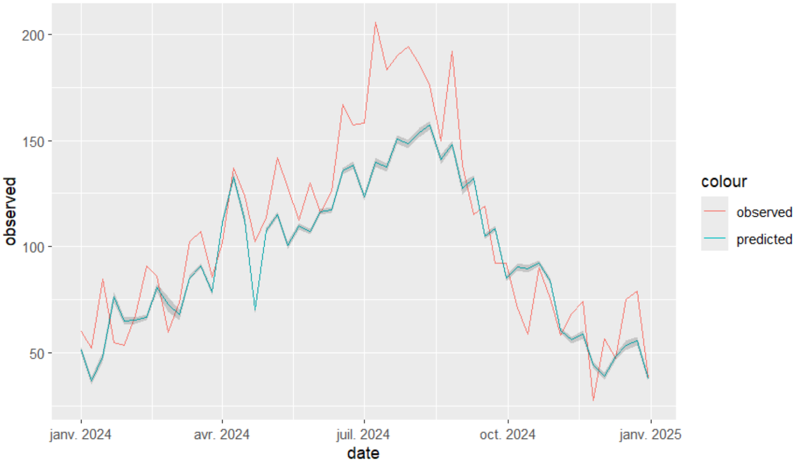
Tail behaviour $\xi \approx 0.002$

- Tail remains light.
- No evidence of heavy-tailed extremes.

Results


PIT histogram
Frequency
0
50
100
150
200
250
0.0
0.2
0.4
0.6
0.8
1.0
PIT

Results



observed
200
150
100
50
janv. 2024
avr. 2024
juil. 2024
oct. 2024
janv. 2025
date
colour
observed
predicted

Bayesian bGEV Model (Stan Implementation)

Regression structure

$$q_i = X_i\boldsymbol{\beta}$$

Non-linear transformation to classical GEV parameters:

$$(\mu_i, \sigma_i) = f(q_i, s, \xi)$$

$$y_i \sim \text{bGEV}(\mu_i, \sigma_i, \xi)$$

Prior specification

$$\beta_j \sim \mathcal{N}(0,1) \quad s \sim \text{LogNormal}(-0.14, 0.29) \quad \xi \sim \mathcal{N}(0, 0.2),\ 0 \leq \xi \leq 0.5$$

Posterior Inference and Model Comparison

Data

- Weekly ozone maxima from 3 monitoring stations
- Same covariates and preprocessing for both methods

Stan estimation

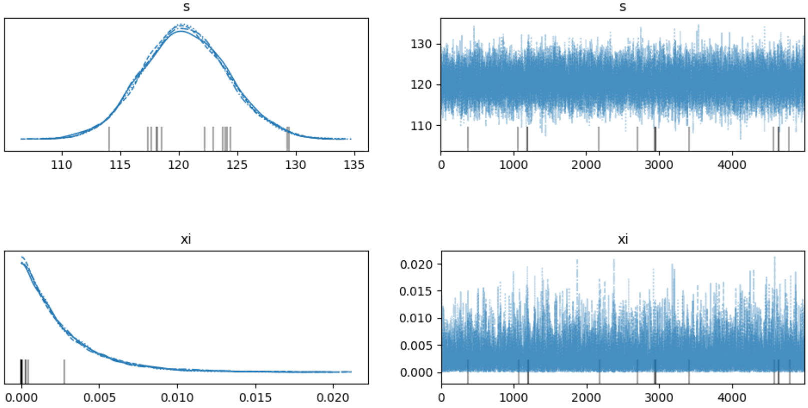
- Hamiltonian Monte Carlo
- Full Bayesian posterior sampling

INLA comparison

- Equivalent bGEV model specification
- Same likelihood and covariates
- Nested Laplace approximation for posterior inference

Objective: assess agreement between full MCMC (Stan) and deterministic approximation (INLA).

STAN posteriors



Stan vs INLA Comparison

Parameter	INLA Mean	Stan Mean
Intercept	10.78	9.357
minTemp	1.46	1.323
maxTemp	1.93	1.156
windSpeed	3.32	0.078
rainSum	-0.02	0.203
s	59.66	120.517
xi	0.0029	0.003

Key observations

- Similar estimates for most regression effects
- Larger discrepancy for spread parameter s
- Shape parameter ξ remains stable

Differences likely due to Laplace approximation (INLA) vs full MCMC (Stan).

Conclusion

Key findings

- **Maximum temperature** is the dominant driver of ozone extremes.
- **Wind speed** and **precipitation** reduce both mean and variability of extremes.
- Tail behaviour is **light** ($\xi \approx 0$, Gumbel-type) across all models.
- Moderate **seasonal peak** in spring–early summer.

Methods

- INLA and Stan yield broadly consistent estimates, validating the Laplace approximation.
- The hybrid model (temporal + heteroskedastic spread) provides the best fit.

Future work

- Model spatial dependence across stations.

References

- Gabriel Escarela. Extreme value modeling for the analysis and prediction of time series of extreme tropospheric ozone levels: A case study. *Journal of the Air Waste Management Association*, 62(6):651–661, 2012. doi:10.1080/10962247.2012.665414.
- Edzer Pebesma and Benedikt Gräler. *Introduction to spatio-temporal variography*, 2025.
- Daniela Castro-Camilo, Raphaël Huser, and Håvard Rue. Practical strategiesfor generalized extreme value-based regression models for extremes.*Environmetrics*, 33(6):e2742, 2022. doi: https://doi.org/10.1002/env.2742. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/env.2742.

Github repository

- https://github.com/alebouvier/BS_project